Xiaoyu Yu (余骁宇)

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Summary

Research focuses on **catalyst design and AOR/OER electrocatalysis**, **AI for Science**, multiphysics interaction and thermoacoustics, 3D ultrasonic reconstruction of gas–liquid interfaces, and high-resolution dynamic optical diagnostics under extreme conditions.

Education

- Ph.D. (Candidate) in Mechanical Engineering., **National University of Singapore** (NUS), Singapore
- M.Sc. in Advancing Theory and Engineering, **Beihang University** (BUAA), China
- B.Sc. (First) in Engineering, **Beihang University** (BUAA), China
- B.Sc. (Second) in Pure and Applied Mathematics, **Beihang University** (BUAA), China

Publications & Patents

▷ Journal & Conference Papers (Selected)

[1] Xiaoyu Yu et al. "Impact of vibrating combustion chamber wall on flame thermoacoustic dynamics". In: *Physics of Fluids* 37.11 (Nov. 2025), p. 114111. DOI: 10.1063/5.0302034. URL: https://doi.org/10.1063/5.0302034.

[2] Xiaoyu Yu, Xiaokang Liu, Jingxuan Li, et al. "Theoretical study on intrinsic thermo-acoustic mode in long flame combustion chamber". In: *China National Symposium on Combustion*. China, 2024.

[3] Xiaoyu Yu, Xiaokang Liu, Jingxuan Li, et al. "The Influence of Lateral Offset of Laminar Premixed Flames on Thermoacoustic Instability of Burners". In: *The International Symposium on Multiscale Simulations of Thermophysics*. China, 2024.

▷ Chinese Invention Patents (Selected)

[4] Jingxuan Li, Xiaoyu Yu, et al. Measurement method for liquid bubble content in the test tank and measurement system for liquid bubble content in the test tank: 202410948864X [P] 2024.07.16

[5] Jingxuan Li, Xiaoyu Yu, et al. Adjustable flow solid nanoparticle generator: 2024102667367 [P] 2024.03.08

[6] Jingxuan Li, Xiaoyu Yu, et al. A measurement method and device for gas-liquid interface of liquid jet in a closed environment: 202510096916X [P] 2025.01.22

Projects

High Frequency 3D Dynamic Characterization of Gas-Liquid Interfaces

— Proposed acoustic scattering measurement theory for high-frequency dynamic microjet and bubble gas-liquid interfaces;

— Independently developed signal inversion algorithm and supporting processing software;

— Built multi-channel high-frequency synchronous ultrasonic array test system, realized millimeter-level bubble measurement accuracy under ultra-low temperature (-196°C / 77K) liquid nitrogen storage tank environment.

Deep Learning-based Reconstruction of Transient Temperature Fields in Thrust Chambers

— Proposed diffusion model (UNet & EDM) fluid field inversion scheme, outperforming traditional neural networks with average reconstruction error lower than 2%.
— Constructed elliptic PDE mesh multi-condition dataset covering 416 chemical working conditions.
— Proposed SSIM combined PAC&I2SB super-resolution evaluation metric, realized high-fidelity transient thermoacoustic reconstruction with 0.327% relative error via DDIM sampling.

Ultrafast Diagnosis of Thermochemical Reactions under Extreme Conditions

— Adopted ultrahigh-speed optical measurement and high spatiotemporal resolution spectroscopy to dynamically capture flame parameters under high temperature and high pressure, revealing ultrafast multiphysics coupling reaction mechanisms.
— Utilized laser interferometry and subpixel edge detection to quantify micro wall vibration, and put forward corresponding vibration suppression control strategies.

Multi-scale 3D PIV Measurement System: Nanoparticle Tracking to High-speed Flow

— Constructed two complete PIV test platforms: high-pressure high-speed PIV (6 MPa, flow velocity 100–600 m/s) and 2D3C stereoscopic PIV system, supporting three-dimensional flow field measurement and high-precision tracking of nanoparticle motion trajectories.

Work Experience

Student Counselor, Beihang University Student Affairs Office

- Responsible for daily office conference arrangement, administrative management and official document drafting work
- Organized large-scale campus events, award ceremonies, sports meetings and freshman military training activities

Chairman, Graduate Student Union of the School of Astronautics

Conducted comprehensive management and coordination of all functional departments; maintained communication and joint research with institutes and technical units.

COVID-19 Campus Isolation Area Volunteer Leader

Served as person-in-charge of campus isolation zone during pandemic prevention and control period.

Skills, Honors & Scholarships

Programming:	Matlab, LabView, Fortran, C, C++, Python, JavaScript, Origin, Chat-GPT
Software:	AutoCAD Professional Qualification, SolidWorks Professional Qualification, MS Office Level 2 (National Computer Rank Examination)
Languages:	IELTS 6.5, CET-6 565 (Top 15% nationwide), Mandarin (Native), Cantonese (Native)

Selected Scholarships: National Scholarship (2025); BUAA Shenyuan Top Scholarship, CATIC Top-class Scholarship, Multiple First-Class Academic Scholarships; Mathematics & Physics Competition Top Scholarship; Freshman Scholarship.

Selected Honors: National University Student Mathematics Competition First Prize (2021,2022); National Undergraduate Graduation Design Competition Second Prize (2023); BUAA Shenyuan Gold Medal (Top 10 students annually, 2025); Outstanding Graduate; Excellent Student Cadre; Top Ten Preacher (2024.4/2024.10/2025.4).

Extracurricular Sports: Badminton Team Champion; Football Runner-up (twice); Basketball Team Champion; Table Tennis, Vocal music.